

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: ARTIFICIAL INTELLIGENCE

COURSE CODE: CSA17

CONSTRAINT-BASED PROBLEM SOLVING

DETAILED PROBLEM STATEMENT

Problem 1: Hospital Doctor Shift Scheduling Using Backtracking Search

Problem Description

- A hospital needs to assign three doctors (D1, D2, D3) to three different shifts: Morning, Afternoon, and Night.
- Each doctor must be assigned exactly one shift.
- Each shift can have only one doctor.
- Certain restrictions are imposed on doctor assignments, making the scheduling process more complex.
- The problem must be solved using the Backtracking Search algorithm.
- The objective is to find a valid schedule that satisfies all constraints without conflicts.

Constraints

- D1 cannot work the Night shift.
- D2 must work before D3.
- D3 cannot work the Morning shift.
- Only one doctor is allowed per shift.
- Every doctor must be assigned exactly one shift.

Problem 2: Robot Navigation in a Grid Environment Using BFS Search

Problem Description

- A robot operates in a 5×5 grid environment.
- The robot starts from the Start (S) position and must reach the Goal (G) position.
- Some grid cells contain obstacles that block movement.
- The robot can move only in four directions: Up, Down, Left, and Right.
- Each movement has a cost of 1 unit.
- The problem requires finding the optimal path while avoiding obstacles.
- Breadth First Search (BFS) must be used to determine the shortest path.

Constraints

- Movement is restricted to four directions.
 - Obstacles cannot be crossed.
 - Every move has a cost of 1.
 - Manhattan Distance is used as a heuristic measure.
 - The shortest valid path must be identified.
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Problem 3: Autonomous Rescue Robot in a Disaster Environment

Problem Description

- An autonomous rescue robot is deployed in a disaster-affected building.
- The robot must locate and reach a survivor positioned at the Goal (G) location.
- The environment contains blocked paths and risky zones.
- The robot has limited sensing capability and can observe only adjacent cells.
- The robot must make decisions dynamically as it explores the environment.
- The solution should minimize both travel cost and risk.
- An appropriate AI search strategy must be used to solve the problem efficiently.

Constraints

- Movement is limited to Up, Down, Left, and Right.
- Obstacles cannot be traversed.
- Only adjacent cells can be observed.
- Each move costs 1 unit.
- Risky zones add an extra cost of 2 units.
- The robot must minimize total cost.
- Knowledge of the environment must be updated dynamically.